PQC DAILY STUDY SERIES

Fault Injection Attacks on PQC Implementations

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Fault Injection Attacks on Post-Quantum Cryptography

Continuing our exploration of physical attacks on Post-Quantum Cryptography (PQC) implementations, today we focus on **Fault Injection Attacks (FIAs)**. While side-channel attacks passively listen to unintended leakages, fault injection attacks are active, aiming to induce errors in a device's operation to extract secret information.

These attacks are a significant threat, especially for PQC algorithms deployed in embedded systems, smart cards, and IoT devices, where an adversary might have physical access.

How do Fault Injection Attacks Work?

An attacker intentionally introduces a fault—a transient or permanent error—into the cryptographic hardware during a sensitive computation. The goal is to disrupt the normal execution flow and analyze the resulting faulty output. By comparing the correct output with the faulty output, an attacker can often deduce parts of the secret key.

- **Methods:** Faults can be induced using various techniques, such as voltage or clock glitching, laser or light pulses, electromagnetic pulses (EM), or temperature variations.
- **Target Operations:** Attackers often target critical operations within PQC algorithms, like polynomial multiplication in lattice-based schemes (ML-KEM, ML-DSA) or hash function computations in hash-based signatures (SLH-DSA).
- **Differential Fault Analysis (DFA):** This is the primary analysis technique used. By injecting a fault and observing the erroneous result, the attacker creates a mathematical relationship that constrains the possible values of the secret key. Repeating this process with different faults can progressively reveal the entire key.

Real-World Applicability: Securing the Edge

Consider a secure firmware update mechanism for an IoT device, signed with ML-DSA. If an attacker can inject a fault into the device's System-on-a-Chip (SoC) during the signature verification process, they might be able to trick the device into accepting a malicious firmware image. This could compromise not just the single device, but potentially an entire network of devices that trust the compromised unit.

Vulnerability of NIST PQC Standards

Research has shown that many PQC algorithms are theoretically vulnerable to FIAs if not implemented carefully. The structured algebraic nature of lattice-based cryptography makes it a prime target.

Fault Injection Vulnerability & Countermeasures

Algorithm Family	Potential Vulnerability	Primary Countermeasure
Lattice-Based (ML-KEM, ML-DSA)	Targeting Number Theoretic Transform (NTT) or polynomial operations can reveal secret coefficients.	**Redundancy & Verification:** Re-computing the operation and comparing results, or verifying output with a checksum before release. Using infection/masking techniques.
Hash-Based (SLH-DSA)	Faults in the hash function can lead to the recovery of parts of the few-time signature (FTS) private key.	**Hardened Hash Cores:** Implementing hash functions with built-in fault detection. State-machine redundancy.
Code-Based (Classic McEliece)	Attacks on the matrix operations or polynomial solvers can leak information about the secret code structure.	**Algorithmic Countermeasures:** Introducing checks and randomized computations to break the deterministic link between fault and key.

Daily Quiz

1. What is the fundamental difference between a side-channel attack and a fault injection attack?
2. Why is Differential Fault Analysis (DFA) effective against cryptographic algorithms?
3. Name one common countermeasure to protect an ML-KEM implementation from a voltage glitch attack.